Study Notes and Quick Reference

ARI5118 - Deep Learning for Computer Vision

Introduction

Object detectors do not natively output a clean list of bounding boxes. A modern detector such as Faster R-CNN[1], YOLO[2], internally evaluates thousands to hundreds of thousands of candidate bounding boxes per image. Many of those boxes pile up around each real object, all with high confidence scores. Without post-processing, an image of a single dog would generate 30+ overlapping "dog" boxes, all technically correct, none useful.

To solve this issue, Non-Maximum Suppression (NMS) is used. It can be described as the post-processing algorithm that turns this redundant pile into one box per object.

Once we have a clean set of detections, we need to know how good they are. To do so, we need a score for hundreds of ranked predictions against a ground truth of a few dozen objects.

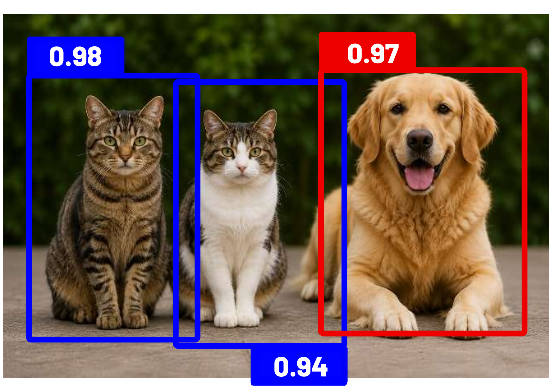
Therefore, for this **Average Precision (AP)** and its multi-class average **mAP (mean Average Precision)** are used.

Fundamentals

Bounding Boxes

Bounding boxes are used in object detection to label an object within an image. Each bounding box has 4 distant parameters.

Table 1: Table summarizing Bounding Boxes Parameters

Sample Image	Parameter	Rationale
 Image generated using ChatGPT. Bounding boxes were added manually for demonstration purposes.	Position + Size	(x, y) In COCO format, this is the top-left corner coordinate plus width and height. In Pascal VOC, it represents the top-left (x_1, y_1) and bottom-right (x_2, y_2) corners.
	Class	The class label as predicted by the object detector. *Example: Dog, Cat*
	Confidence Score	The probability that the bounding box contains the object as indicated by the class label. Value between 0 and 1.

Intersection over Union (IoU)

General Information

Intersection over Union (IoU) is a metric in computer vision that measures how much a predicted region like a bounding box overlaps with the ground truth region, defined as the area of their intersection divided by the area of their union. IoU values range from 0 (no overlap) to 1 (perfect overlap) and are commonly used to decide whether a detection is correct during model evaluation.

For bounding boxes A and B, IoU is defined as follows

$$\mathrm{IoU}(A, B) = \frac{\mathrm{area}(A \cap B)}{\mathrm{area}(A \cup B)}$$

Properties of IoU:

- **Symmetric.** IoU$(A, B) =$ IoU(B, A).
- **Bounded in** $[0, 1]$**.** Always.

Worked IoU Calculation

Let Bounding Box A have corners (50, 50) and (150, 150), width 100, height 100, therefore the area is10,000. Let Bounding Box B have corners (100, 100) and (200, 200), width 100, height 100, therefore the area is10,000.

Intersection rectangle has corners $(\max(50{,}100), \max(50{,}100)) = (100{,}100)$ and $(\min(150{,}200), \min(150{,}200)) = (150{,}150)$.

- Intersection area = $(150 - 100) \times (150 - 100) = 50 \times 50 = 2{,}500$.
- Union area = $10{,}000 + 10{,}000 - 2{,}500 = 17{,}500$.
- IoU = $2{,}500/17{,}500 \approx 0.143$.

Non-Maximum Suppression

As described before, a vision object detector does not output one tidy box per object. It outputs thousands of candidate boxes, many of which pile up on top of each other around the same real object.

Standard NMS

The algorithm takes three inputs: a set of candidate boxes, their confidence scores, and an IoU threshold that defines the overlap. The algorithms works as follows and is done for each class independently:

1. Find the box with the highest confidence score. Move it to the final output list where this box is kept.
2. Compare every remaining box against the kept box. If the IoU above the threshold, delete that box, it is a presumed duplicate.
3. Repeat from step 1 on whatever boxes remain, until the pool of the pre-NMS outputs is empty.

The following observations could be made about this algorithm:

Greedy - The algorithm always picks the current highest-scoring survivor. It never looks ahead, never reconsiders a decision, and does not attempt a globally optimal selection.

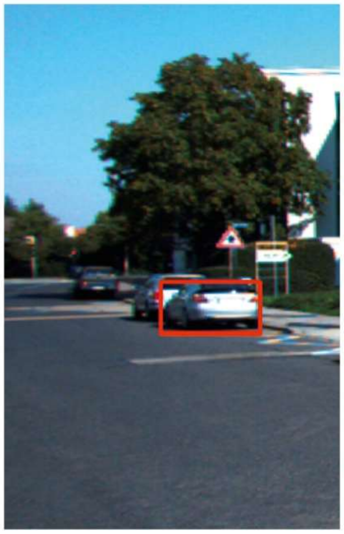
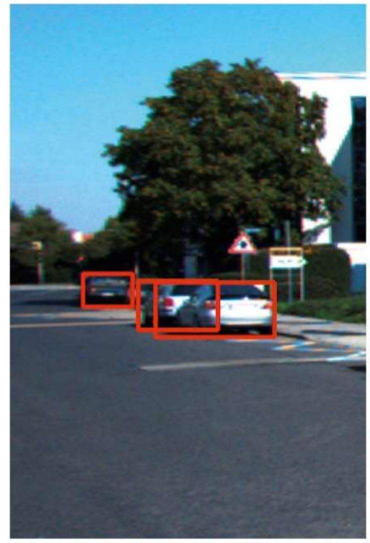
Hard threshold. The suppression decision is binary. The bounding box either stays untouched or is deleted entirely. There is no middle ground, no partial penalty. This is the property Soft-NMS later relaxes.

Per-class. Standard practice runs NMS independently for each object class. A "dog" box and a "cat" box sitting on top of each other will both survive, because they are processed in separate runs. Only boxes of the same class compete with each other.

Failure case of NMS

Standard NMS has a fundamental flaw as shown in Table 1. Standard NMS sees the lower-scoring car box, finds it has IoU greater than the threshold with the higher-scoring box, and *deletes it*.

Table 2: Comparison of NMS to Soft-NMS[3]

	
NMS	Soft-NMS

To clean the output results as shown above, NMS is applied.

The first step is to eliminate all the bounding boxes that have a threshold

Soft-NMS

Bodla. et. al {Citation}made the observation that making the hard binary decision and deleting it may not be the most viable option. Instead, the score of an overlapping box shall be decayed proportional to how much it overlaps. If the overlapping box covers a real second object, its decayed but still keeps it which may end up a true positive at the end. If the overlap is strong enough it will fall below the confidence threshold.

Soft-NMS is still a greedy algorithm as it processes boxes in score order and never reconsiders an earlier decision.

Linear Soft-NMS

The confidence

$$s_i = \begin{cases} s_i \text{if } \mathrm{IoU}(M, b_i) < N_t \\ s_i \cdot \left(1 - \mathrm{IoU}(M, b_i)\right) \text{if } \mathrm{IoU}(M, b_i) \geq N_t \end{cases}$$

Gaussian Soft-NMS

$$s_i = s_i \cdot \exp\left(-\frac{\mathrm{IoU}(M, b_i)^2}{\sigma}\right) \quad \forall b_i \in B$$

Example

Refer to the walkthrough.ipynb. Example is provided here for both NMS and Soft-NMS

Detection Evaluation

Once a detector produces a clean, ranked list of detections, it needs to be scored against the ground truth. This section builds up the Average Precision (AP)/mAP concepts.

True Positive, False Positive, False Negative

Each predicted boundary box, at evaluation time, is one of:

- **True Positive (TP):** correct class as per ground-truth box of the same class.
- **False Positive (FP):** prediction does not match ground-truth or else it was already taken within the same class.
- **False Negative (FN):** a ground-truth box that no prediction matched.

Precision and Recall

In the following section, Precision and Recall are defined.

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \qquad Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

- **Precision**: Of the boxes you submitted, what fraction were right?
- **Recall**: Of the objects that exist, what fraction did you find?

Average Precision

11-point interpolated AP

Sample recall at 11 evenl -spaced points $\{0.0, 0.1, 0.2, \ldots, 1.0\}$. At each recall r, take the maximum precision observed at any recall ≥ r. This is the interpolated precision, it smooths the wiggles in the raw PR curve by taking the monotone envelope from the right. It is defined in the equation below.

$$\mathrm{AP} = \frac{1}{11} \sum_{r \in \{0, 0.1, \ldots, 1.0\}} p_{\mathrm{interp}}(r), \qquad p_{\mathrm{interp}}(r) = \max_{\tilde{r} \geq r} p(\tilde{r})$$

This is the form Everingham et al. [4] define in the VOC challenge paper.

Example

An example of this is extracted from [5] technical post which shows a good example step by step. Here is the output of the bounding boxes.

They are ranked based on confidence, the second column is the prediction with IoU.

Rank	Correct?	Precision	Recall
1	True	1.0	0.2
2	True	1.0	0.4
3	False	0.67	0.4
4	False	0.5	0.4
5	False	0.4	0.4
6	True	0.5	0.6
7	True	0.57	0.8
8	False	0.5	0.8
9	False	0.44	0.8
10	True	0.5	1.0

For each particular row, the AP the precision and recall where calculated. For example for first item Precision is TP = 1/1 therefore it is 1.0, however Recall is 1/5 = 0.2. 5 is the total number of predictions possible.

Rank	Correct?	Precision	Recall
1	True	1.0 ↑	0.2 ↑
2	True	1.0 –	0.4 ↑
3	False	0.67 ↓	0.4 –
4	False	0.5 ↓	0.4 –
5	False	0.4 ↓	0.4 –
6	True	0.5 ↑	0.6 ↑
7	True	0.57 ↑	0.8 ↑

The following items are calculated.

The results are plotted on a curve as shown below.

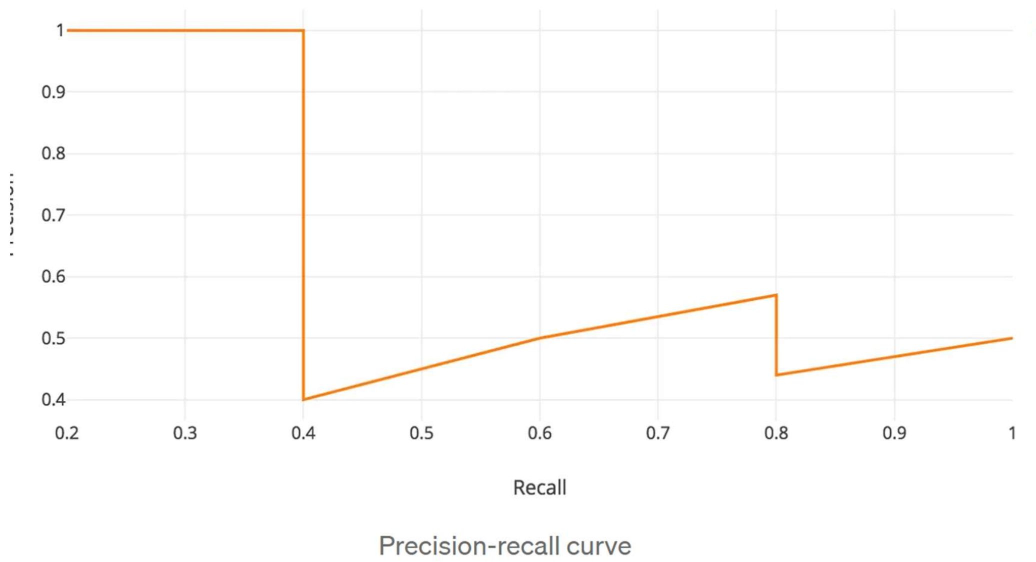


Figure 1: Precision-Recall Curve

The average decision is defined by the following equation.

$$AP = \int_0^1 \mathrm{p(r)dr}$$

The interpolated average precision would be a smoothed version of this as shown where the interpolated average precision is a smoothed version of the precision-recall curve where precision at each recall level is replaced by the maximum precision achieved at any recall level greater than or equal to that point. The interpolation version is used to eliminate non-monotonic behavior of the raw precision-recall curve, therefore the output metric is interpolated

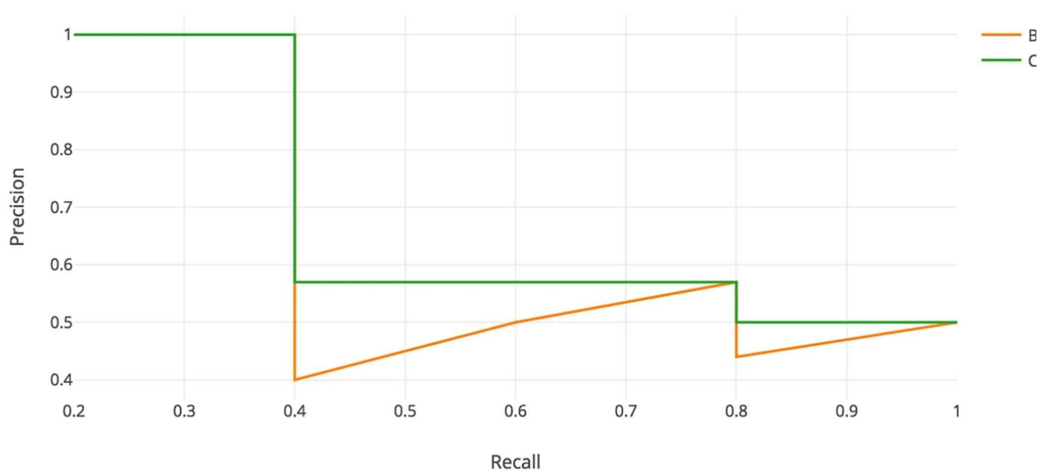


Figure 2: Precision-Recall Curve with Interpolation

The interpolation can be calculated as follows.

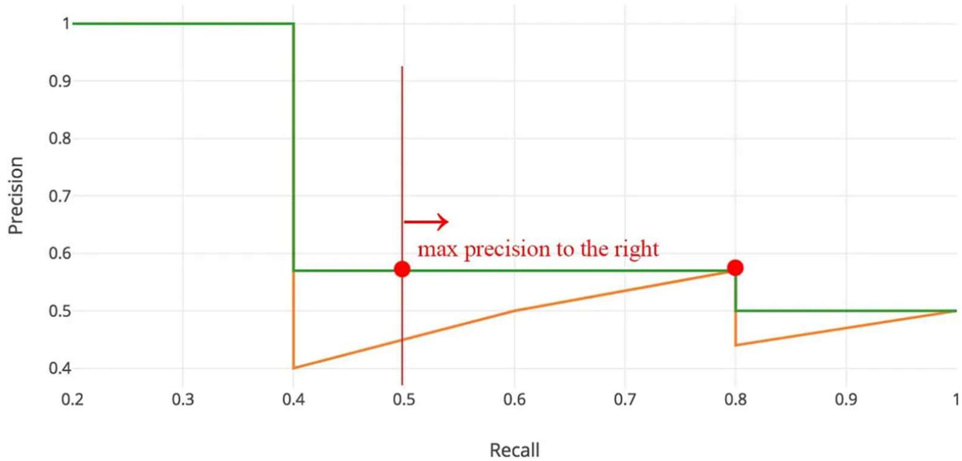


Figure 3: Average precision calculation example

The 11-point interpolated value by taking recall at (0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0) is

$$\frac{1}{11}\sum_{r\in\{0,0.1,\ldots,1\}}\max_{\tilde{r}\geq r} p(\tilde{r})$$

The average precision at each point is calculated as shown below.

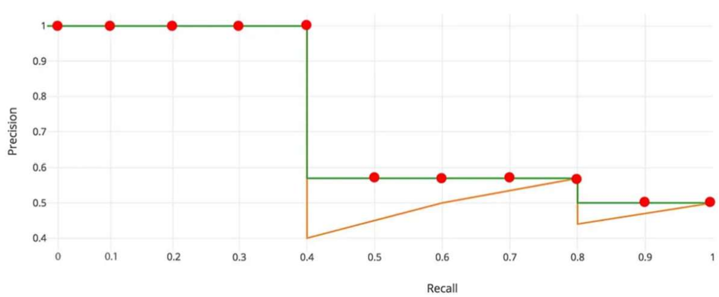


Figure 4: 11-point interpolated AP

$$\boldsymbol{AP} = \frac{1}{11} \text{x} \,(1 + 1 + 1 + 1 + 1 + (0.57 \text{ x } 4) + (0.5 \text{ x } 2)) = 0.75$$

High-Level Summary

All topics can be combined together as follows. The image shows what is the output of the raw detector. Eventually is passed through a score-threshold which discards items that do not meet the criteria. Eventually, the Non-Maximum Suppression algorithm can take different forms. Finally the cleaned output is evaluated using Average Precision. The results of all the Average Precision classes are used to calculate the Mean AP. A full image of this is shown below.

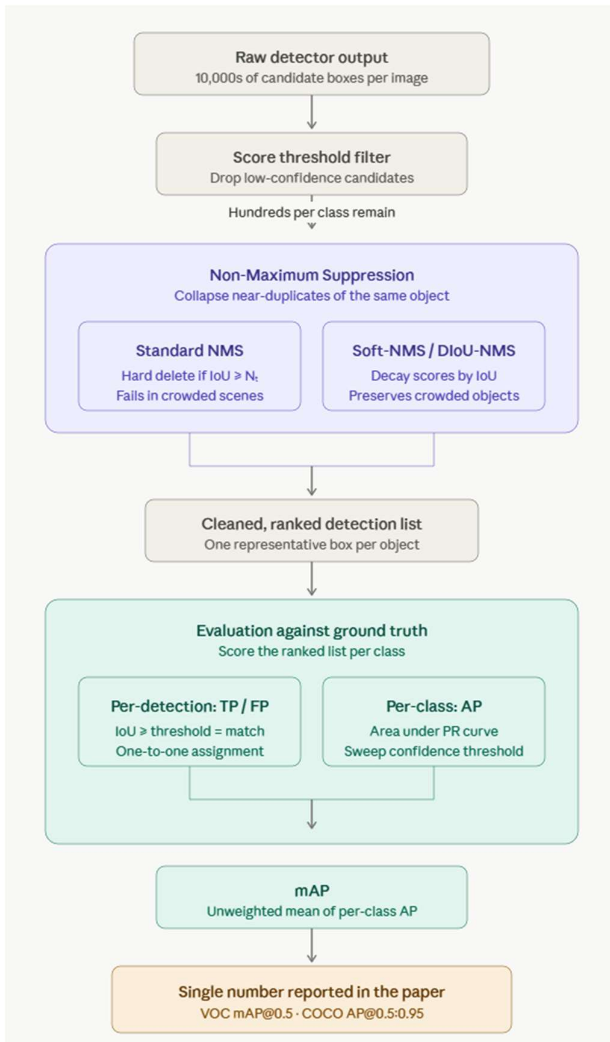


Figure 5: Summary of all this. (Image generated by Claude Opus 4.7)

Analogy

The analogy of NMS can be compared to the following. Imagine a wedding photographer reviewing the day's shots after a photoshoot. The photographer has hundred of photos of the couple. There are group of photos and withing each group, the photos would be very similar as they are taken in a short period of time, however they may have some slight variations of that moment. For example, some slight head turns, micro-expressions, and eyes half-closed. The photographer's role is to pick one photo per moment. The selected photo shall be the best representative of that moment and after that the photographers shall discard the near-duplicates.

That is what NMS does for bounding boxes. Boxes whose overlap with a higher-scoring box exceeds a threshold are near-duplicates of the same moment and get suppressed going back to the wedding analogy.

Now imagine the photographer has to grade his own selection of photos. The photographer would extract the following metrics.

- **Precision**: of the photos kept, what fraction were genuinely good?
- **Recall**: of the moments worth capturing, what fraction were preserved?

The photographer might keep a tight selection (high precision, lower recall) or a generous one (lower precision, higher recall). The trade-off depends on the operating point. Average Precision integrates the photographer's performance across all operating points. The results shall answer the following question. How good is the photographer overall, regardless of how strict the selection rule was?"

Mean Average Precision is the same number averaged across the 20 wedding events the photographer does shot this year. This is used as a per-category fairness check that prevents the bride-and-groom photos from dominating the score.

Figure 6: Image generated using ChatGPT based on the above analogy

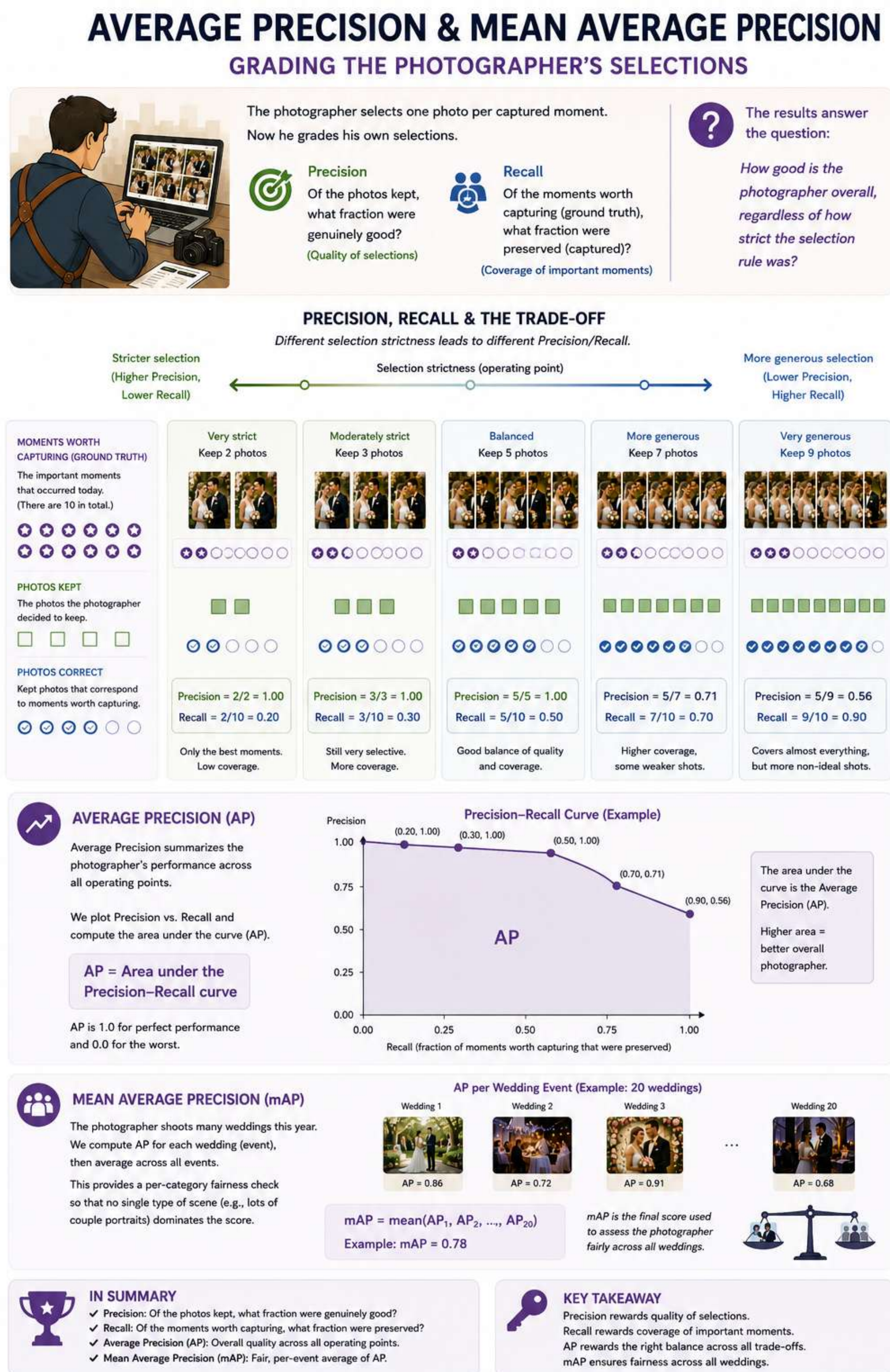


Figure 7: Image generated using ChatGPT based on the above analogy

Real-world examples and applications of the technique

Visual object detection models are at the core of many modern intelligent systems. They are used in autonomous vehicles that must simultaneously identify pedestrians, cyclists, traffic signs, and other vehicles across camera, LiDAR, and Radar sensor streams[6] . Moreover, they are used in medical imaging, where deep learning systems localize tumors and lesions in MRI and CT scans with life-critical precision. In the vision detection, NMS plays a critical role at inference time. For example, in an autonomous vehicle pipeline, multiple overlapping bounding boxes are generated around the same pedestrian. NMS collapses these into a single, most-confident prediction before the vehicle's planning system acts on them. The quality of these detections is then quantified using AP, which captures the trade-off between precision and recall across different confidence thresholds. Standard benchmarks like KITTI, nuScenes, and Waymo evaluate models precisely this way, reporting per-class AP for cars, pedestrians, and cyclists to validate deployment readiness. Together, NMS and AP form an inseparable pair in the object detection pipeline: NMS ensures clean, deployment-ready outputs at inference time, while AP validates during development that the model meets the accuracy standards demanded by real-world, safety-critical use.

Quick Reference

Equations

Quantity	Formula
IoU	$\mathrm{IoU}(A,B) = \dfrac{\mathrm{area}(A \cap B)}{\mathrm{area}(A \cup B)}$
Precision	$P = \dfrac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$
Recall	$R = \dfrac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$
F1	$\mathrm{F}_1 = \frac{2\mathrm{PR}}{\mathrm{P+R}}$
Linear Soft-NMS rescore	$s_i \leftarrow s_i \cdot (1 - \mathrm{IoU}) \quad \text{if IoU} \geq N_t$
Gaussian Soft-NMS	$s_i \leftarrow s_i \cdot \exp\left(-\dfrac{\mathrm{IoU}^2}{\sigma}\right)$
Average Precision	$AP = \int_0^1 \mathrm{p(r)dr}$
11-point AP	$\dfrac{1}{11} \sum_{r \in \{0, 0.1, \ldots, 1\}} \max_{\tilde{r} \geq r} p(\tilde{r})$
All-point AP	$\sum_k (r_{k+1} - r_k) \cdot \max_{\tilde{r} \geq r_{k+1}} p(\tilde{r})$
mAP	$\frac{1}{C} \sum_{c=1}^{C} AP_c$
COCO AP	mean of AP over IoU ∈ {0.50, 0.55, …, 0.95} averaged over classes

Parameter Ranges

Parameter	Typical	Rationale
VOC IoU threshold for TP	0.5	Set by VOC Standard
COCO IoU thresholds for AP	{0.50, …, 0.95} step 0.05	10 thresholds, average
Pre-NMS confidence threshold (greedy NMS)	0.01 – 0.05	Drops obviously-wrong boxes *before* NMS runs. Lower values keep more candidates for NMS to process; higher values speed up NMS but risk losing marginal TPs. Faster R-CNN default is 0.05; YOLO variants often use 0.25.
Post-NMS confidence	0.3 – 0.5	Applied *after* NMS to decide which surviving boxes are shown to the user or submitted for evaluation. Only affects

threshold (greedy NMS)		visualisation / deployment; does not affect AP computation (AP sweeps all thresholds internally
Post-rescoring score threshold (Soft-NMS)	10^{-4} to 10^{-2} (0.0001 – 0.01)	The decisive parameter. Applied after Soft-NMS finishes decaying scores. Without it, tens of thousands of near-zero-score boxes survive. Bodla et al. use 10^{-4} in their experiments and report 0.005 s per image at 10^{-2}.
Max detections per image (COCO evaluation)	100	The COCO evaluation server selects the top 100 detections per image. Bodla et al. note that reducing from 400 to 100 costs ~0.1% AP.

Misconceptions Catalogue — NMS & Detection Evaluation Metrics

NMS Threshold

Misconception: "Setting the IoU threshold to 0 disables NMS / keeps every box."

Reality: IoU = 0 means any positive overlap triggers suppression. This is the most aggressive setting, not the least. Disjoint boxes survive, but anything that touches a higher-scoring box dies.

Misconception: "A high IoU threshold (e.g. 0.9) suppresses more aggressively."

Reality: Opposite. A high threshold means only near-identical boxes get suppressed. Most duplicates pass through, flooding the output with false positives.

Misconception: "IoU threshold 0.0 means only one box survives in the entire image."

Reality: Two non-overlapping boxes have IoU = 0, which is not greater than 0, so both survive. Non-overlapping boxes are never suppressed regardless of the threshold.

Confusing the Two Thresholds

Misconception: "The confidence threshold and the IoU threshold do the same job inside greedy NMS."

Reality: They do entirely different things. The confidence threshold is a pre-filter that decides which boxes enter NMS. The IoU threshold decides which boxes get suppressed inside NMS.

Soft-NMS

Misconception: "Soft-NMS removes boxes like standard NMS, just more gently."

Reality: Soft-NMS never removes a single box. It only decays scores. Boxes leave the final output only when a post-rescoring confidence threshold prunes them.

Misconception: "The σ parameter is what controls what survives in Soft-NMS."

Reality: σ controls the *shape* of the decay curve, therefore how steeply scores drop with IoU. But a box decayed to 0.0001 still exists in the output unless a final score threshold removes it. The decisive parameter is the post-rescoring threshold, not σ.

Misconception: "Soft-NMS requires no hyperparameters."

Reality: The paper's abstract claims this, but in practice Soft-NMS trades the IoU threshold for the Gaussian parameter σ, and adds a mandatory post-rescoring score threshold.

Misconception: "Soft-NMS is faster than standard NMS because it avoids deletion."

Reality: Same $O(N^2)$ complexity. Soft-NMS replaces the deletion operation with a multiplication.

Misconception: "Soft-NMS helps everywhere equally."

Reality: Soft-NMS helps most in crowded same-class scenes where standard NMS over-suppresses legitimate overlapping objects. In sparse scenes with isolated objects, the two algorithms behave nearly identically.

Precision, Recall, and AP

Misconception: "AP and accuracy measure roughly the same thing."

Reality: Accuracy requires True Negatives, which are undefined in detection as the negative class is every empty patch of pixels in every image which is not enumerable. However, AP operates in precision-recall space, where TN plays no role.

Misconception: "If I multiply all confidence scores by 0.1, AP decreases."

Reality: AP depends on the rank order of detections, not their absolute scores. Any monotone-preserving rescaling leaves the order unchanged and therefore leaves AP unchanged.

Misconception: "A second predicted box matching an already-matched ground truth is ignored."

Reality: It is counted as a False Positive. Each ground truth can be matched at most once; duplicates damage precision. If duplicates were ignored, a detector could game AP by replicating predictions.

VOC vs COCO

Misconception: "VOC mAP and COCO mAP are comparable numbers."

Reality: VOC mAP uses a single IoU threshold of 0.5. COCO's headline "AP" averages over 10 IoU thresholds {0.50, 0.55, ..., 0.95}. A model reporting 0.42 on COCO may be stronger than one reporting 0.75 on VOC. COCO's AP50 is the metric directly comparable to VOC mAP.

Key Papers

In this section, 4 papers are described, each contributing to above mentioned topics and concepts.

Histograms of Oriented Gradients for Human Detection

Navneet Dalal and Bill Triggs

CVPR 2005

https://ieeexplore.ieee.org/document/1467360

Summary

Dalal et al [7]. introduced two contributions as it will be discussed in the following paragraph. The first, the Histogram of Oriented Gradients (HOG) feature descriptor, is combined with a linear SVM classifier for pedestrian detection. Eventually, over time this has been largely superceeded. The second, a methodological choice in the post processing step, is that the established greedy max-suppression is using a fixed IoU threshold for bounding-box object detection. This approach replaces the clustering-and-averaging approach of Viola and Jones [8]. This approach has been used in modern detector's inference pipeline.

Contribution

The idea of suppressing non-maximal responses had been around in image processing since Rosenfeld and Thurston [9]. Moreover, the idea of deduplicating overlapping bounding-box detections had been used in Viola and Jones face detection [8], but they did it by clustering overlapping boxes into disjoint subsets and averaging their coordinates, not by suppressing non-maxima. Dalal et. al. was that the proposed algorithm when faced with overlapping bounding-box detections from a multi-scale sliding-window pipeline, the boxes are processed in descending score order, keeping the maximum, and discard anything that overlaps it above a fixed IoU threshold. They also showed empirically that this scheme outperformed Viola-Jones style averaging on the INRIA pedestrian benchmark. In effect, they imported the concept of non-maximum suppression from edge-detection, where it had existed for decadies into bounding-box object detection, and demonstrated that it beat the prior alternative.

PASCAL Visual Object Classes (VOC) Challenge

Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, Andrew Zisserman

International Journal of Computer Vision (IJCV), Vol. 88, pp. 303–338, 2010

https://link.springer.com/article/10.1007/s11263-009-0275-4

Summary

This paper [4] introduces a benchmarking landmark when it comes to visual object recognition and detection. It provides standardized annotations, and evaluation procedures such that different teams can compare there detectors onto equal metrics. This paper is relevant to this topic as it formalizes the 11-point interpolation method Average Precision (AP) and mean AP (mAP) as standard object detection metrics.

Key Contribution

The key contributions are summarized in the table below.

Table 3: Table showcasing the key contributions of VOC Challenge

Topic	**VOC Paper's contribution**
IoU Threshold	VOC defined IoU ≥ 0.5 as the standard match criterion for a true positive detection.
Precision-Recall curves	VOC suggests PR curves as the evaluation output; defined the 11-point interpolation method

Improving Object Detection With One Line of Code

Navaneeth Bodla, Bharat Singh, Rama Chellappa, Larry S. Davis

ICCV 2017

https://ieeexplore.ieee.org/document/8237855

Summary

Bodla et al. [10] identify a fundamental flaw in the greedy NMS, the hard suppression of overlapping boxes causes true detections to be missed when multiple objects occupy the same region. Soft-NMS replaces the binary suppression step with a continuous score decay function, either linear or Gaussian. The decay would be proportional to the IoU overlap with the highest-scoring box, so nearby detections are penalised rather than eliminated. The method requires no retraining and shares the same $O(N^2)$ complexity as standard NMS. It yields consistent mAP gains of 1.1–1.7% on PASCAL VOC and MS-COCO across Faster-RCNN, R-FCN, and Deformable-RFCN detectors.

Key Contribution

The key contribution of this paper is that Soft-NMS can replace hard suppression in NMS with a gradual score decay, so overlapping boxes are penalized instead of immediately discarded, which helps detect close or crowded objects more reliably. In the context of NMS, that means it preserves useful detections that standard NMS might wrongly remove, improving object-detection accuracy with essentially no extra training or computational cost.

Microsoft COCO: Common Objects in Context

Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, Piotr Dollar

Computer Vision – ECCV 2014

https://link.springer.com/chapter/10.1007/978-3-319-10602-1_48

Summary

This paper [11] introduces the MS COCO dataset and benchmark. It is consists of 91 categories, 328,000 images, 2.5 million labelled instances with per-instance segmentation masks. This paper introduces COCO's evaluation protocol averages AP over 10 IoU thresholds {0.50, 0.55, ..., 0.95}. The authors argue that the single-IoU criterion popularized by PASCAL VOC can reward detectors that merely ‘roughly’ localize objects, whereas COCO’s multi-IoU averaging explicitly penalizes such loose localization and therefore better reflects true localization quality.

Key Contribution

The COCO paper is important because it changed how object detectors are evaluated. Instead of measuring AP at a single IoU (like VOC’s 0.5), it averages AP over many IoU thresholds from 0.5 to 0.95, so detectors must both find objects and localize them tightly to score well. This became the standard benchmark.

References

[1] S. Ren, K. He, R. Girshick, and J. Sun, 'Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks', arXiv.org. Accessed: May 21, 2026. [Online]. Available: https://arxiv.org/abs/1506.01497v3

[2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, 'You Only Look Once: Unified, Real-Time Object Detection', in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA: IEEE, Jun. 2016, pp. 779–788. doi: 10.1109/CVPR.2016.91.

[3] N. Hoanh, 'Improving Faster R-CNN Framework for Fast Vehicle Detection', *Math. Probl. Eng.*, vol. 2019, pp. 1–11, Nov. 2019, doi: 10.1155/2019/3808064.

[4] S. P. H. Boroujeni, N. Mehrabi, H. Alzorgan, M. Fazeli, and A. Razi, 'All You Need for Object Detection: From Pixels, Points, and Prompts to Next-Gen Fusion and Multimodal LLMs/VLMs in Autonomous Vehicles', *Image Vis. Comput.*, vol. 169, p. 105944, May 2026, doi: 10.1016/j.imavis.2026.105944.

[5] J. Hui, 'mAP (mean Average Precision) for Object Detection', Medium. Accessed: May 24, 2026. [Online]. Available: https://jonathan-hui.medium.com/map-mean-average-precision-for-object-detection-45c121a31173

[6] S. P. H. Boroujeni, N. Mehrabi, H. Alzorgan, M. Fazeli, and A. Razi, 'All You Need for Object Detection: From Pixels, Points, and Prompts to Next-Gen Fusion and Multimodal LLMs/VLMs in Autonomous Vehicles', *Image Vis. Comput.*, vol. 169, p. 105944, May 2026, doi: 10.1016/j.imavis.2026.105944.

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